

Global Rainfall Map in Near Real Time (GSMaP_NRT) Data Format Description

This document describes data format and information of Global Rainfall Map in Near Real Time (hereafter refers as GSMaP_NRT) distributed from JAXA Global Rainfall Watch, which was developed based on activities of the GSMaP (Global Satellite Mapping of Precipitation) project. The GSMaP project is promoted for the study "Production of a high-precision, high-resolution global precipitation map using satellite data," sponsored by Core Research for Evolutional Science and Technology (CREST) of the Japan Science and Technology Agency (JST).

1. Product Overview

Table 1 Summary of GSMaP_NRT Products

No	Parameter [unit]	Data format	Coverage	Grid size	Horizontal resolution	Temporal resolution	FTP directory
1	Hourly Rain Rate [mm/h]	4-byte float plain binary, little-endian	Global (60°N-60°S)	3600 x 1200	0.1 degree grid box	Hourly	Latest 24-hr: /realtime/latest/ Archive: /realtime/archive/ YYYY/MM/DD/
2	Satellite Information Flag	4-byte singed integer plain binary, little-endian					/realtime/sateinfo /b12222/
3	Observation Time Flag	4-byte float plain binary, little-endian					/realtime/timeinfo /b12222/
4	Hourly Rain Rate in text format (old) [mm/h]	ASCII, CSV format	Global but divided to 54 areas	200 rows x 400 lines			/realtime/txt/AAA BBB/YYY/MM/ DD/
5	Hourly Rain Rate in text format (new) [mm/h]	ASCII, CSV format	Divided to 15 areas	--			/realtime/txt/XX_ ZZZZZ/YYY/M M/DD/
6	Daily Rainfall [mm/h]	4-byte float plain binary, little-endian	Global (60°N-60°S)	1440 x 480	0.25 degree grid box	Daily (averaged from 00Z to 23Z of the specified day)	/realtime/daily/00 Z-23Z/YYYMM/
7						Daily (averaged from 12Z of previous day to 11Z of the specified day)	/realtime/daily/p1 2Z-11Z/YYYMM /

Note: **YYYY**: 4-digit year, **MM**: 2-digit month, **DD**: 2-digit day, **AAA**: latitude of the corner of left-top position (2-digit latitude + S or N), **BBB**: longitude of the corner of left-top position (3-digit longitude + E or W), and **XX_ZZZZZ**: area name (9-digit).

2. Hourly Rainfall and Flag Files in Binary (products (1)-(3))

2.1. Basic Information

Temporal resolution:	1 hour (hourly data)
Grid resolution:	0.1 degrees latitude/longitude grid (10km at the equator). Latitude and longitude of the first grid [1, 1] is [59.95°N, 0.05°E].
Domain:	Global (60°N-60°S)
Data latency:	4 hours after observation
Data archived period:	File will be removed when reanalysis data (GSMaP_MVK) is uploaded (about 3days).

2.2. FTP Directory Information

Hourly Rain Rate data;	
Latest 24 hour data:	/realtime/latest/
Archive:	/realtime/archive/YYYY/MM/DD/
Satellite Information Flag;	/realtime/sateinfo/YYYY/MM/DD/
Observation Time Flag;	/realtime/timeinfo/YYYY/MM/DD/

where;

YYYY: 4-digit year;
MM: 2-digit month; and
DD: 2-digit day.

2.3. File Naming Rules

Data and flag files are named according to the following rules;

Hourly Rain Rate data:	gsmmap_nrt.YYYYMMDD.HHNN.dat
Satellite Information Flag:	gsmmap_nrt.YYYYMMDD.HHNN.sateinfo.dat
Observation Time Flag:	gsmmap_nrt.YYYYMMDD.HHNN.timeinfo.dat

where;

YYYY: 4-digit year;
MM: 2-digit month;
DD: 2-digit day;
HH: 2-digit hour; and
NN: 2-digit minute (currently fixed as 00).

2.4. Data Format

Data format was slightly changed since 10 Oct. 2008 due to algorithm version up. Satellite Information Flag and Observation Time Flag are newly added after 10 Oct. 2008. History of version up is described in GSMaP_NRT_HISTORY.txt file in ftp server.

All binary files are produced in little-endian byte order platform, and archived with compressed using “gzip”. Grid of those files consists of 3600 rows x 1200 lines, which are longitude-latitude elements corresponding to a 0.1 x 0.1 degree grid that covers the global region from 60°N to 60°S. The center longitude and latitude of the first pixel [1, 1] (left top corner) are [0.05°E, 59.95°N] (Figure 1).

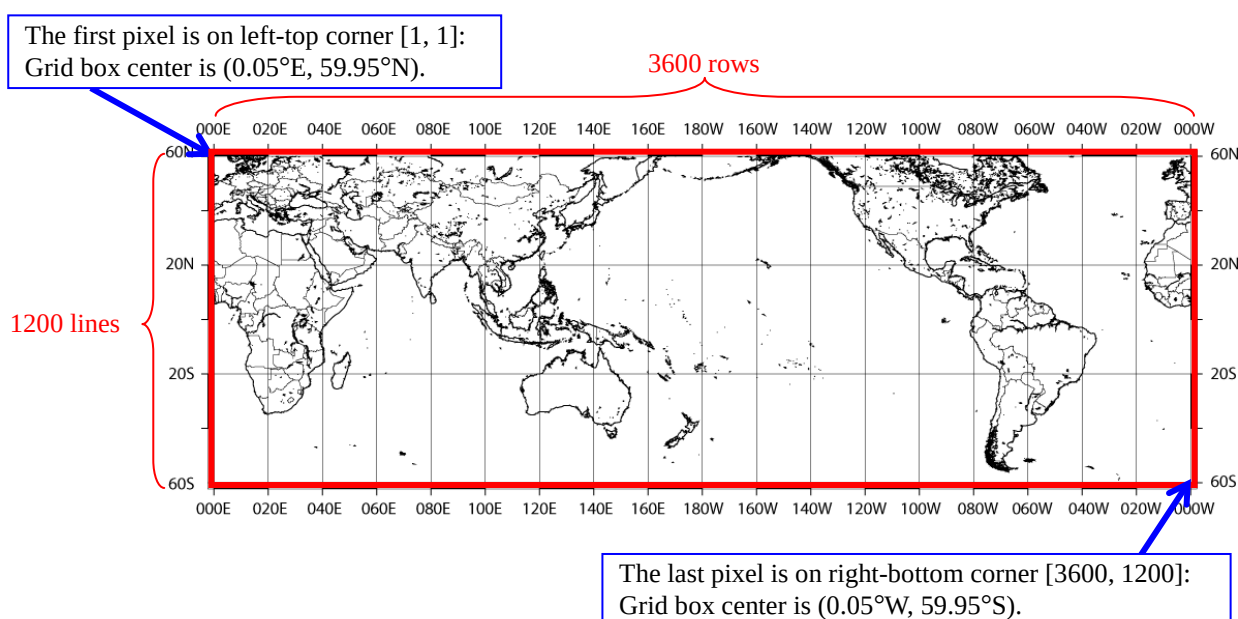


Figure 1 Data Coverage Map (Rain and Flag data)

2.5. Stored Values of Hourly Rain Rate

“Hourly Rain Rate” data are stored in 4-byte float plain binary format. Unit is [mm/hr]. Negative value denotes missing in observation data or no retrieval was done in microwave retrieval algorithm. Detailed description for missing data is shown in Table 2.

Table 2 Stored Values of Hourly Rain Rate

Value	Description
(positive)	Hourly rain rate [mm/hr].
-4	Missing due to sea ice in microwave retrieval algorithm.
-8	Missing due to low temperature in microwave retrieval algorithm.
-99	Missing due to no observation by IR and/or microwave.

Please note that specification of missing value has changed since 10 October 2008. Products prior to 10 October 2008, no missing value is defined (all set to zero).

2.6. Stored Values of Satellite Information Flag

“Satellite Information Flag” data are stored in 4-byte integer plain binary format. Satellite and sensor name are assigned to each bit, and the flag indicates all satellite/sensor which are used in estimation of rainfall at each pixel during one-hour time period. If the flag shows 0, there is no satellite observation by both microwave and geo-stationary IR. If flag shows negative value, there is NO microwave radiometer observation. Following meanings are assigned to each bit in 32-bit integer (Table 3).

Table 3 Stored Values of Satellite Information Flag

Pixel Value		Description	
Value	Bit	Sensor Category	Satellite/Sensor
1	0	Microwave imager and/or sounder aboard low orbital satellite	TRMM/TMI
2	1		Aqua/AMSR-E
4	2		DMSP-F13/SSM/I
8	3		DMSP-F14/SSM/I
16	4		DMSP-F15/SSM/I
32	5		DMSP-F16/SSMIS
64	6		DMSP-F17/SSMIS
128	7		NOAA-19/AMSU-A/MHS
256	8		MetOp-A/AMSU-A/MHS
512	9		DMSP-F18/SSMIS
1024	10		GCOM-W/AMSR2
2048	11		GPM-Core/GMI
4096	12		NOAA-18/AMSU-A/MHS
8192	13		MetOp-B/AMSU-A/MHS
16384-32768	14-15		not used
65536	16	Infrared Imager aboard Geo-stationary meteorological satellite (before 22Z 28 Mar. 2012)	GOES-EAST
131072	17		GOES-WEST
262144	18		INDEX
524288	19		METEOSAT
1048576	20		MTSAT
2097152– 536870912	21–29		not used
1073741824	30	Infrared Imager aboard Geo-stationary meteorological satellite (since 23Z 28 Mar. 2012)	
–(negative)	31	No microwave radiometer observation	

2.7. Stored Values of Observation Time Flag

“Observation Time Information Flag” are in 4-byte float plain binary format. The Flag indicates relative time of latest microwave radiometer observation at each pixel, and 0 means start time of the file (**HH** in file name). Values are stored as indicated in Table 4.

Table 4 Stored Values of Observation Time Flag

Value	Description
$0 \leq X < 1$	If value is positive, microwave radiometer observation is available at the pixel during current one-hour period. X ($0 \leq X < 1$) indicates relative observation time of latest microwave radiometer, and is stored as differences from the start time of the file. For example, if UTC of the file (HH) = "01" and $X = 0.2$, observation time of the pixel will be 01:12 UTC.
$X \leq 0$	If value is negative, NO microwave radiometer observation is available at the pixel during time period of the file. X ($X \leq 0$) indicates relative observation time of latest microwave radiometer, and stored as differences from the start time of the file. For example, if UTC of the file (HH) = "01" and $X = -2.5$, latest observation time of microwave radiometer at the pixel will be 22:30 UTC of previous day.
$X = -999$	No microwave observation (Missing)

2.8. GrADS Control File

Sample control files of the Grid Analysis and Display System (GrADS) for each product are also available from ftp server.

Hourly Rain Rate data: /realtime/archive/GSMaP_NRT.hourly.rain.ctl
 Satellite Information Flag: /realtime/sateinfo/GSMaP_NRT.hourly.sat.ctl
 Observation Time Flag: /realtime/timeinfo/GSMaP_NRT.hourly.time.ctl

About usage of GrADS tool, please see GrADS home page (<http://grads.iges.org/grads/head.html>).

2.9. File Size

Approximately 800 Kbyte (with gzip), and 17 Mbyte (uncompress) for each file.

3. Hourly rainfall in text format (old) (product (4))

3.1. Basic Information

Temporal resolution: 1 hour (hourly data)
 Grid resolution: 0.1 degrees latitude/longitude grid (10km at the equator).
 Domain: Global (60°N-60°S), but data are divided into 54 subset area files (area of 40 degree for latitude, 20 degree for longitude).
 Data latency: 4 hours after observation
 Data archived period: Stored only one week on the ftp server.

3.2. FTP Directory Information

Data files are archived at following directory; /realtime/txt/**AAABBBB/YYYY/MM/DD/**

where;

YYYY: 4-digit year;

MM: 2-digit month;

DD: 2-digit day;

AAA: The corner of left-top position is appeared with Latitude (2-degit latitude + S or N); and

BBBB: The corner of left-top position is appeared with Longitude (3-degit longitude + E or W).

3.3. File Naming Rules

Data files are named according to following rules; `gsmmap_nrt.YYYYMMDD.HHNN.AAABBBB.csv`

where;

YYYY: 4-digit year;

MM: 2-digit month;

DD: 2-digit day;

HH: 2-digit hour;

NN: 2-digit minute (currently fixed as 00);

AAA: The corner of left-top position is appeared with Latitude (2-degit latitude + S or N); and

BBBB: The corner of left-top position is appeared with Longitude (3-degit longitude + E or W).

3.4. Data Format

Text files are stored in CSV format. Unit is [mm/hr]. Negative value denotes missing in observation data or no retrieval same as binary format data (see Table 2).

Each file is one of global coverage fractionated 54 areas and consists of 200 rows x 400 lines which is longitude-latitude elements corresponding to a 0.1 x 0.1 degree grid that covers each fractionated area. The number of effective digits is zero pint two digits. This file is available to open using Microsoft Excel directory. Figure 2 is example of data coverage for the case of **AAABBBB** = 60N140E.

List of area code **AAABBBB** and its corresponding latitude and longitude of left-top and right-bottom pixels are stored in following file; `/realtime/txt/area_list.pdf`.

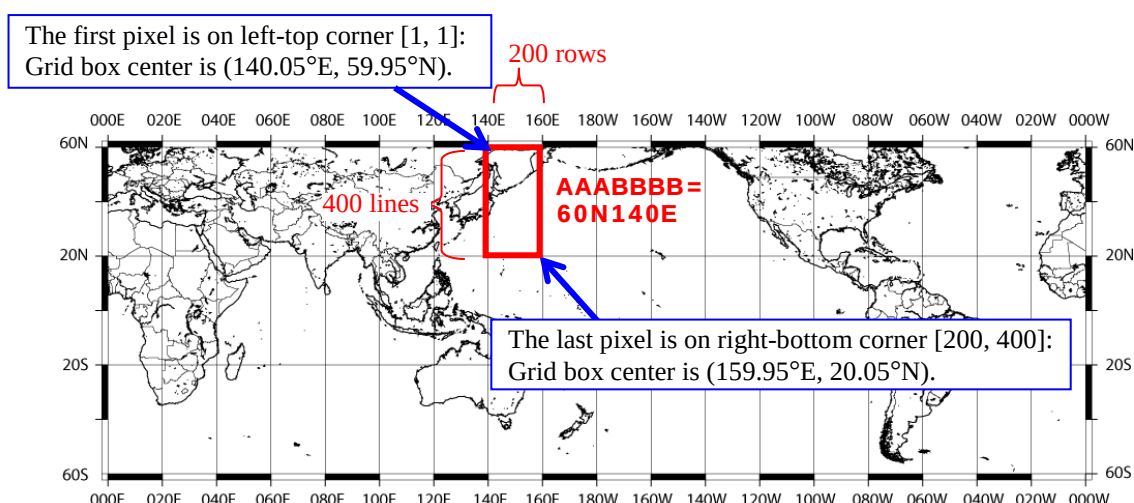


Figure 2 Example of Data Coverage (Text file)

3.5. File Size

Approximately 5 Kbyte (with gzip), and 560 Kbyte (uncompress) for each file.

4. Hourly rainfall in text format (new) (product (5))

4.1. Basic Information

Temporal resolution:	1 hour (hourly data)
Grid resolution:	0.1 degrees latitude/longitude grid (10km at the equator).
Domain:	15 areas (see 4.4)
Data latency:	4 hours after observation
Data archive period:	File will be removed when reanalysis data (GSMaP_MVK) is uploaded (about 3days).

4.2. FTP Directory Information

Data files are archived at following directories;

Hourly Data; /realtime/txt/ **XX_ZZZZZZ/YYYY/MM/DD/**

where;

YYYY: 4-digit year;
MM: 2-digit month;
DD: 2-digit hour; and
XX_ZZZZZZ: 9-digit area name.

4.3. File Naming Rules

Data files are named according to following rules;

Hourly Data;
gsmap_nrt.**YYYYMMDD_HH00_XX_ZZZZZZ.csv**

where;

YYYY: 4-digit year;
MM: 2-digit month;
DD: 2-digit day;
HH: 2-digit hour; and
XX_ZZZZZZ: 9-digit area name.

4.4. Area definition in text format

15 areas are defined for Text format as in Figure 3.

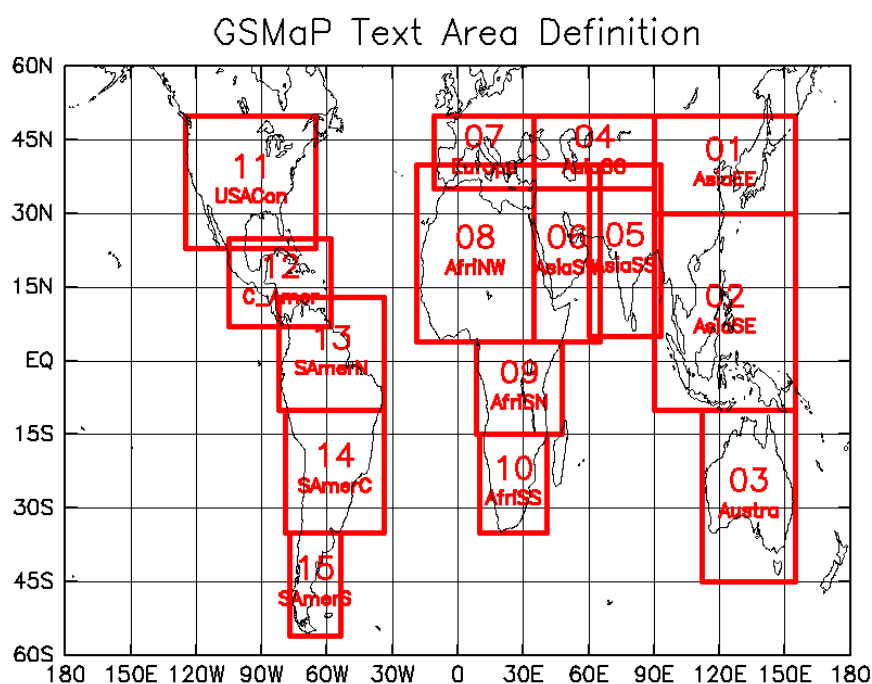


Figure 3 Definition of new Text Area

Table 5 Corner latitude and longitude of each area

Area name	Longitude (W)	Longitude (E)	Latitude (S)	Latitude (N)	Description
01_AsiaEE	90	155	30	50	East Asia
02_AsiaSE	90	155	-10	30	South East Asia
03_Austra	112	155	-45	-10	Australia
04_AsiaCC	35	90	35	50	Central Asia
05_AsiaSS	60	93	5	40	South Asia
06_AsiaSW	35	65	4	40	Arabian Peninsula and East Africa
07_Europe	-11	35	35	50	Europe
08_AfriNW	-19	35	4	40	North West and Central Africa
09_AfriSN	8.5	48	-15	4	Southern Africa (North)
10_AfriSS	10	41	-35	-15	Southern Africa (South)
11_USACon	-125	-65	23	50	USA (Contiguous)
12_C_Amer	-105	-58	7	25	Central America
13_SAmerN	-82	-34	-10	13	South America (North)
14_SAmerC	-79	-34	-35	-10	South America (Central)
15_SAmerS	-77	-54	-56	-35	South America (South)

4.5. Data Format

Text files are stored in CSV format (see Figure 4). Unit is [mm/hr]. Data with missing value are omitted. All text files are archived with compressed using “zip”.

This data format is available in the ArcGIS (ESRI ArcMap 10.0), verified by Mr. Fujioka (ICHARM).

Lat	Lon	RainRate
49.95	89.95	0
49.85	89.95	0
49.75	89.95	0
49.65	89.95	0
...	...	...

Figure 4 Example of text format

4.6. File Size

Approximately 200 Kbyte (with zip), and 1.6 Mbyte (uncompress) for each file.

5. Daily rainfall (products (6)-(7))

5.1. Basic Information

Temporal resolution: 24 hours average (daily data)
 Two definition of “daily”:
 00Z-23Z average: from 00Z to 23Z of the day
 12Z-11Z average: from 12Z of the previous day to 11Z of the day
 Grid resolution: 0.25 degrees latitude/longitude grid (25km at the equator)
 Domain: Global (60°N-60°S)
 Data latency: 4 hours after the end of accumulation period
 Data archived period: File will be removed when reanalysis data (GSMaP_MVK) is uploaded (about 3days).

5.2. FTP Directory Information

Data files are archived at following directories;

Daily data (00Z-23Z average);

/realtime/daily/00Z-23Z/YYYYMM/

Daily data (12Z-11Z average);

/realtime/daily/p12Z-11Z/YYYYMM/

where;

YYYY: 4-digit year; and

MM: 2-digit month.

5.3. File Naming Rules

Data files are named according to following rules;

Daily data (00Z-23Z average);

gsmap_nrt.**YYYYMMDD**.0.25d.daily.00Z-23Z.dat

Daily data (12Z-11Z average);

gsmap_nrt.**YYYYMMDD**.0.25d.daily.p12Z-11Z.dat

where;

YYYY: 4-digit year;

MM: 2-digit month; and

DD: 2-digit day.

5.4. Data Format

All binary files are produced in little-endian byte order platform, and archived with compressed using “gzip”. Unit is [mm/hr]. Missing value is -999.9.

Grid of those files consists of 1440 x 480 pixels, which are longitude-latitude elements corresponding to a 0.25 x 0.25 degree grid that covers the global region from 60°N to 60°S. The center longitude and latitude of the first pixel [1, 1] (left top corner) is [0.125°E, 59.875°N].

5.5. GrADS Control File

Sample control files of the Grid Analysis and Display System (GrADS) for each product are also available from ftp server.

Daily data (00Z-23Z average);

`/realtime/daily/00Z-23Z/GSMaP_NRT.daily.00Z-23Z.ctl`

Daily data (12Z-11Z average);

`/realtime/daily/p12Z-11Z/GSMaP_NRT.daily.p12Z-11Z.ctl`

About usage of GrADS tool, please see GrADS home page (<http://grads.iges.org/grads/head.html>).

5.6. File Size

Approximately 800 Kbyte (with gzip), and 2.7 Mbyte (uncompress) for each file.

6. Algorithm and references

6.1. Algorithm

The dataset of “Global Rainfall Map in Near Real Time” is near-real-time version of GSMaP algorithm. Details of the latest GSMaP algorithm are described in following documents and references in Section 6.2.

- Global Satellite Mapping of Precipitation (GSMaP) for GPM: Algorithm Theoretical Basis Document (ATBD)” (http://www.eorc.jaxa.jp/GPM/doc/algorithm_e.htm).

6.2. References

Papers describing the GSMaP project and algorithm are as follows.

(About GSMaP project)

- K. Okamoto, T. Iguchi, N. Takahashi, K. Iwanami and T. Ushio, 2005: The global satellite mapping of precipitation (GSMaP) project. *25th IGARSS Proceedings*, 3414-3416.
- K. Okamoto, T. Iguchi, N. Takahashi, T. Ushio, J. Awaka, S. Shige, and T. Kubota, 2007: High precision and high resolution global precipitation map from satellite data. *ISAP 2007 Proceedings*, 506-509.
- T. Kubota, S. Shige, H. Hashizume, K. Aonashi, N. Takahashi, S. Seto, M. Hirose, Y. N. Takayabu, K. Nakagawa, K. Iwanami, T. Ushio, M. Kachi, and K. Okamoto, 2007: Global Precipitation Map using

Satelliteborne Microwave Radiometers by the GSMaP Project : Production and Validation. *IEEE Trans. Geosci. Remote Sens.*, **45(7)**, 2259-2275.

(About microwave imager algorithm)

- K. Aonashi, J. Awaka, M. Hirose, T. Kozu, T. Kubota, G. Liu, S. Shige, S., Kida, S. Seto, N. Takahashi, and Y. N. Takayabu, 2009: GSMaP passive, microwave precipitation retrieval algorithm: Algorithm description and validation. *J. Meteor. Soc. Japan*, **87A**, 119-136.
- A. Taniguchi, S. Shige, M. K. Yamamoto, T. Mega, S. Kida, T. Kubota, M. Kachi, T. Ushio, and K. Aonashi, 2013: Improvement of high-resolution satellite rainfall product for Typhoon Morakot (2009) over Taiwan. *J. Hydrometeor.*, **14**, 1859-1871.
- S. Shige, M.K. Yamamoto, and A. Taniguchi, 2014. Improvement of TMI rain retrieval over the Indian Subcontinent. *Geophys. Monogr. Ser.* (in print).
- M.K. Yamamoto, and S. Shige, 2014: Implementation of an orographic/nonorographic rainfall classification scheme in the GSMaP algorithm for microwave radiometers. *Atmos. Res.* (in print)

(About microwave sounder algorithm)

- S. Shige, T. Yamamoto, T. Tsukiyama, S. Kida, H. Ashiwake, T. Kubota, S. Seto, K. Aonashi and K. Okamoto, 2009: The GSMaP precipitation retrieval algorithm for microwave sounders. Part I: Over-ocean algorithm. *IEEE Trans. Geosci. Remote Sens.*, **47**, 3084-3097.
- S. Kida, S. Shige, and T. Manabe, 2010: Comparison of rain fractions over tropical and sub-tropical ocean obtained from precipitation retrieval algorithms for microwave sounders. *J. Geophys. Res.*, **115**, D24101, doi:10.1029/2010JD014279.
- S. Kida, T. Kubota, M. Kachi, S. Shige, and R. Oki, 2012: Development of precipitation retrieval algorithm over land for a satellite-borne microwave sounder. *Proc. of IGARSS 2012*, 342-345.

(About microwave imager/sounder algorithm)

- T. Kubota, S. Shige, M. Kachi, and K. Aonashi. 2011: Development of SSMIS rain retrieval algorithm in the GSMaP project. *Proc 28th ISTS*, 2011-n-46.

(About microwave-IR combined algorithm)

- T. Ushio, T. Kubota, S. Shige, K. Okamoto, K. Aonashi, T. Inoue, N., Takahashi, T. Iguchi, M. Kachi, R. Oki, T. Morimoto, and Z. Kawasaki, 2009: A Kalman filter approach to the Global Satellite Mapping of Precipitation (GSMaP) from combined passive microwave and infrared radiometric data. *J. Meteor. Soc. Japan*, **87A**, 137-151.
- T. Ushio, T. Tashima, T. Kubota, and M. Kachi, 2013: Gauge Adjusted Global Satellite Mapping of Precipitation (GSMaP_Gauge), *Proc. 29th ISTS*, 2013-n-48.

(About NRT system)

- M. Kachi, T. Kubota, T. Ushio, S. Shige, S. Kida, K. Aonashi, and K. Okamoto, 2011: Development and utilization of “JAXA Global Rainfall Watch” system. *IEEJ Transactions on Fundamentals and Materials*, **131**, 729-737. (In Japanese with English abstract)
- T. Ushio, and M. Kachi, 2009: Kalman filtering application for the Global Satellite Mapping of Precipitation (GSMaP). *Chapter for “Satellite Rainfall Applications for Surface Hydrology” (Edited by Mekonnen Gebremichael and Faisal Hossain)*, Springer, ISBN978-9048129140, 105-123.

Additional related papers are listed on the JST/CREST GSMaP Project Website

http://sharaku.eorc.jaxa.jp/GSMaP_crest/html/publications.html

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